

statement: 'The bad packets stop here'. Press **Enter** at the **Boot** prompt. Select the desired language; an information dialogue box tells you that pressing **Cancel** will reboot the system. Continue by clicking **OK**. Select the type of keyboard and the time zone, then enter the correct date and time, if required. Select the hard disk on which to install IPCop. Beware that all the data on this disk will be erased. Then click **OK** to continue. Here, a very interesting screen greets you, irrespective of whether the installation is on the hard disk or on a flash drive. Select the desired disk type from **Flash/Disk**. The installer will create the required file systems and swap space, ask whether you have an older backup to be restored, and complete the installation.

Reboot the system and continue to configure the box. Enter the host name and domain name, before selecting the Red (Internet) interface type and Ethernet card. Note that here you can configure any Ethernet card to any interface. In the earlier version, the first detected Ethernet was always assigned to Green, and it was tricky to change it later. Also, to identify the card easily, you can set it in 'LED blinking on' mode. Assign Green, Red and Blue (if required) cards, and enter the Green IP address and subnet mask. Also, select the Red type, such as Static, PPPoE, etc. You may enable DHCP if desired, and enter root, admin and backup passwords to complete this part of the configuration.

IPCop configuration

Reboot the system and wait till you see the login screen (which can be used to log in as the root user). Use a client computer browser to open <https://IPCopIP:8443> and authenticate yourself as the **admin** with an earlier defined password. The first step is to complete the set-up by configuring the Internet. This trial set-up uses PPPoE on the Red interface. Proceed to **Network > Dialup**, and enter the PPPoE username and password. From the **Home** screen (**System > Home**), click the **Connect** button to connect to the Internet. Configure the IPCop Green IP as the default gateway and DNS in the client TCP/IP settings.

The next important step is to upgrade to the latest release. Start by checking whether new updates are available; go to the **System > Update** page. Select **Refresh Update List** and check for the availability of new updates. Do apply all the available patches. After the updates, IPCop will be upgraded to the latest version 2.0.6 (as of July 1, 2013).

Configure client systems under Green and Blue with the corresponding IPCop interface IP address as their default gateway and DNS. An external DNS IP address can also be used.

That is all... you can now start browsing securely.

To conclude this article, let me answer a few questions

that are frequently asked by users at various organisations, and explain three real life scenarios where a properly configured firewall could have helped. Let me emphasise that not just a branded, but any properly configured firewall would have saved the day.

Firewall FAQs

"All our systems have anti-virus. Do we require a network firewall?" Yes, anti-virus is not a substitute for a network firewall.

"All our systems have a host-based firewall. Do we require a network firewall?"

Yes. First, configuring host-based firewalls on multiple systems itself is a big task. Also, these firewalls sit only on operating systems and do not protect other network devices such as network printers, access points, Network Attached Storage (NAS) boxes, etc.

"Since we have a firewall costing Rs 100,000 installed in our network, aren't our systems protected?"

Not sure. While auditing, I have observed that these firewalls are not configured properly. The best approach to configuring any firewall is to deny all traffic from the firewall and allow only the desired traffic by creating rules, one by one. Such an approach generates a lot of reconfiguration calls after installation is completed and hence is not preferred by many. You may buy or install any good firewall. Much depends on its configuration rather than anything else.

"Is it possible to filter https traffic using IPCop?"

Yes, let me try to explain. When you browse a http website, all data travels in clear text format. In simple language, this means an intruder can sniff this traffic and read through your login ID and password. To overcome this, https websites use an asymmetric encryption mechanism so that the traffic is sent in encrypted form from the browser to the Web server, thus ensuring its confidentiality. Now, since this traffic is encrypted, even a firewall can't read it! Then the question is, how can it be filtered? There are two ways:

- Blocking the https protocol will disable all https traffic—even all required https websites will be blocked. You can allow various sites by finding their IP addresses and creating corresponding 'allow' rules. This method is tedious, specially considering that many websites have multiple IP ranges, and a few website IP addresses change.
- Blocking all direct https traffic and allowing it only via Advanced Proxy (inbuilt in IPCop). Here, all browsers will have to be configured with proxy settings. The https proxy functions as follows:
 1. Browser https traffic is terminated by the IPCop proxy (instead of the destination Web server).
 2. The URL filter reads the requested URL.